Stock Price Prediction

A Case Study of Safaricom: Using historical stock data and advanced supervised machine learning algorithms to forecast price movements and support informed investment decisions in the Kenyan market.

Executive Summary

Leveraging supervised machine learning to forecast Safaricom stock prices with improved accuracy over traditional methods.

1. Project Focus

This project focuses on forecasting Safaricom stock prices using supervised machine learning algorithms. By analyzing historical stock data patterns, we aim to develop predictive models that support informed investment decisions in the Kenyan equity market. The data-driven approach leverages years of trading history to identify trends and price movements.

1. Expected Outcome

The primary outcome is improved prediction accuracy compared to traditional forecasting methods such as moving averages and basic statistical models. By employing advanced ML techniques including Random Forest, XGBoost, and LSTM networks, we anticipate delivering reliable short-term price forecasts that empower investors with actionable insights for the Safaricom stock.

15%

Accuracy Gain

Problem Statement

Stock market volatility creates significant challenges for accurate price prediction. Traditional forecasting methods such as technical analysis and fundamental analysis have demonstrated limited accuracy in capturing market dynamics. Furthermore, there is a notable lack of tailored predictive tools specifically designed for Kenyan markets like the Nairobi Securities Exchange. This gap highlights the critical need for reliable short-term stock price forecasting solutions.

Market Volatility

Limited accuracy in traditional methods

Solution

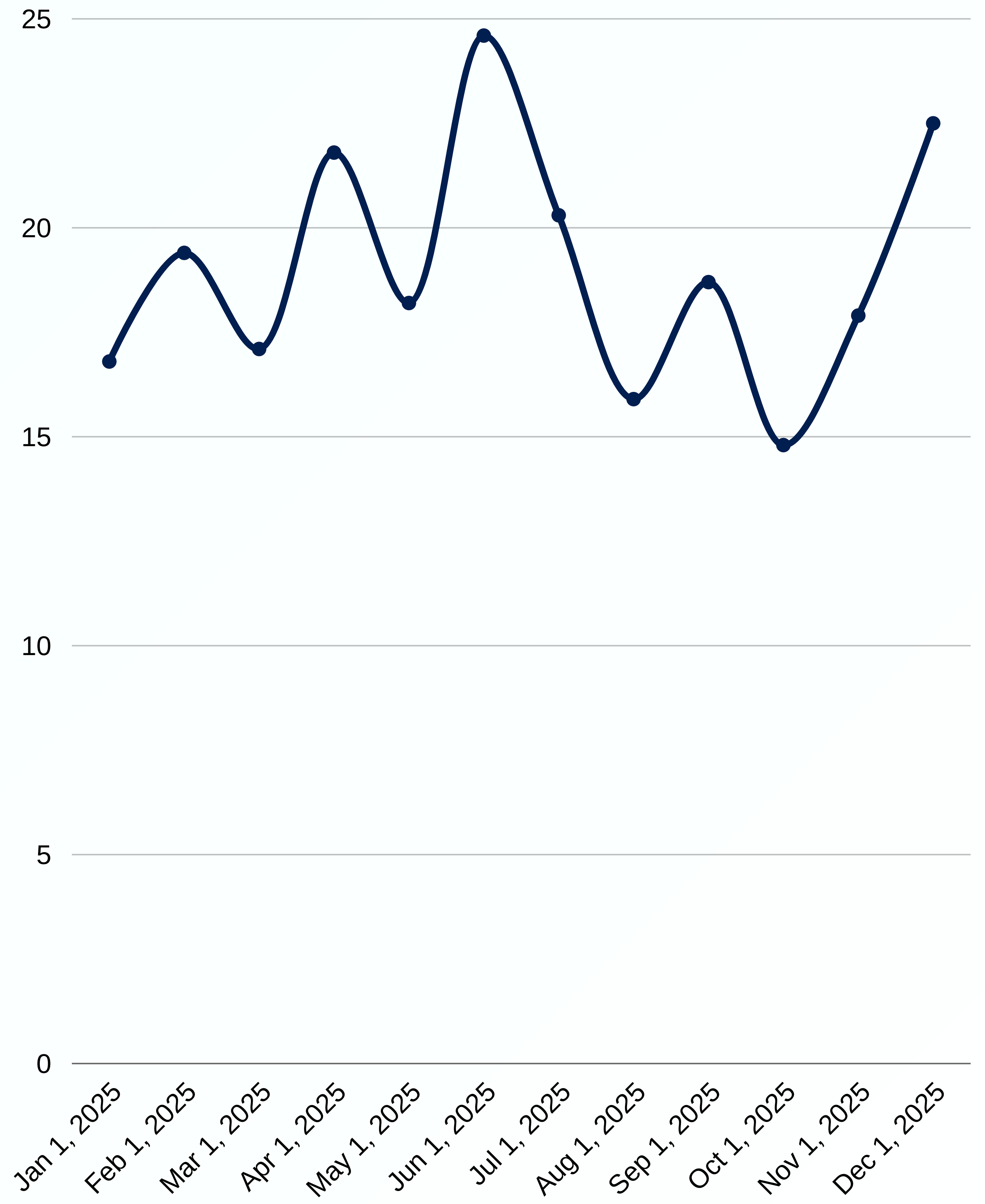
Data-driven ML approach

Tool Gap

Lack of localized predictive tools for Kenyan equities

Forecast Need

Short-term forecasts needed



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Project Objectives

Primary objective: Predict Safaricom stock prices accurately using supervised machine learning techniques to support data-driven investment decisions.

Forecast Horizon

Develop accurate short-term stock price predictions within 1 to 30 day forecast windows, enabling timely investment decisions for Safaricom equity traders and portfolio managers.

Trend Analysis

Identify and analyze underlying price trends and patterns in historical Safaricom stock data to understand market behavior and improve prediction reliability.

Model Comparison

Systematically compare multiple supervised learning algorithms including Linear Regression, Random Forest, XGBoost, and LSTM to determine the optimal approach for Kenyan market conditions.

Visualization

Develop intuitive visualization tools and dashboards for results interpretation, enabling stakeholders to easily understand predictions and make informed investment decisions.

1. Data Source

Kaggle dataset containing Safaricom stock market data. Publicly available and well-documented source ensuring reproducibility. Historical range covers 5 to 10 years of daily trading records, providing sufficient data points for robust model training and validation.

1. Key Features

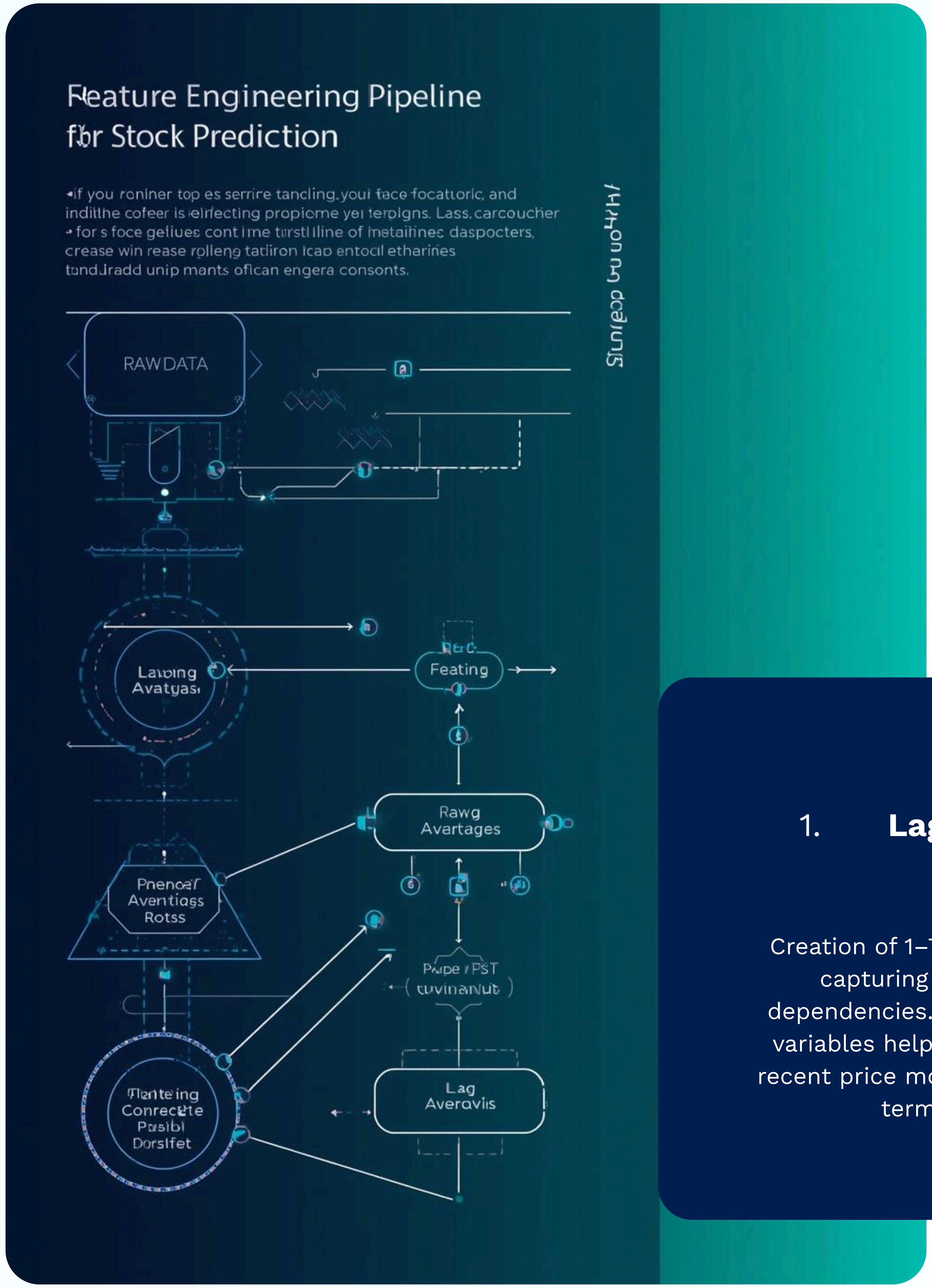
Primary features include Open, High, Low, and Close prices for each trading day. Trading Volume captures market activity levels. Date field enables time-series analysis and temporal feature engineering for trend identification.

1. Data Quality

Preprocessing requirements include handling missing values and outliers. Data normalization ensures consistent model input. Quality checks verify data integrity across the historical range. Clean, structured data is essential for accurate stock price predictions.

Dataset Overview

Our analysis leverages comprehensive historical stock data from Kaggle, spanning 5-10 years of daily Safaricom trading records. This robust dataset provides the foundation for accurate supervised machine learning predictions.



Feature Engineering

Transforming Raw Data

Feature engineering transforms raw stock data into meaningful predictive inputs. We create derived variables that capture temporal patterns, price momentum, and market volatility to enhance model performance.

1. Lag Features

Creation of 1–7 day delay features capturing historical price dependencies. These time-shifted variables help models learn from recent price movements and short-term patterns.

1. Moving Averages

Calculation of 7, 14, and 30-day moving average windows to smooth price fluctuations and identify underlying trends. These indicators reveal momentum and support/resistance levels.

1. Technical Indicators

RSI (Relative Strength Index) and MACD (Moving Average Convergence Divergence) for momentum analysis. Plus volatility measures and time-based features like day of week and month.

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Methodology Overview

A systematic approach to building accurate stock price prediction models through structured data science workflows.

1. Data & Analysis

Data Preprocessing: Clean raw stock data by handling missing values, removing outliers, and normalizing price scales for consistent model input.
Exploratory Data Analysis: Identify patterns, seasonality, and correlations in Safaricom stock data through statistical analysis and visualization techniques.
Feature Engineering: Create lag features, moving averages, and technical indicators like RSI and MACD to enhance prediction accuracy.

1. Modeling & Deployment

Model Building: Train supervised learning algorithms including Linear Regression, Random Forest, XGBoost, and LSTM networks on engineered features.
Model Evaluation: Assess performance using RMSE, MAE, MAPE, and R^2 metrics with cross-validation and backtesting on historical data splits.
Deployment Plan: Implement real-time prediction pipeline with automated data ingestion and dashboard visualization for investor decision support.

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6 Key Steps

1. **Linear Regression**

Serves as the baseline model for comparison. Simple, interpretable approach that assumes linear relationships between features and target variable. Provides a benchmark against which more complex models are measured. Fast training and easy to understand coefficients.

1. **Random Forest**

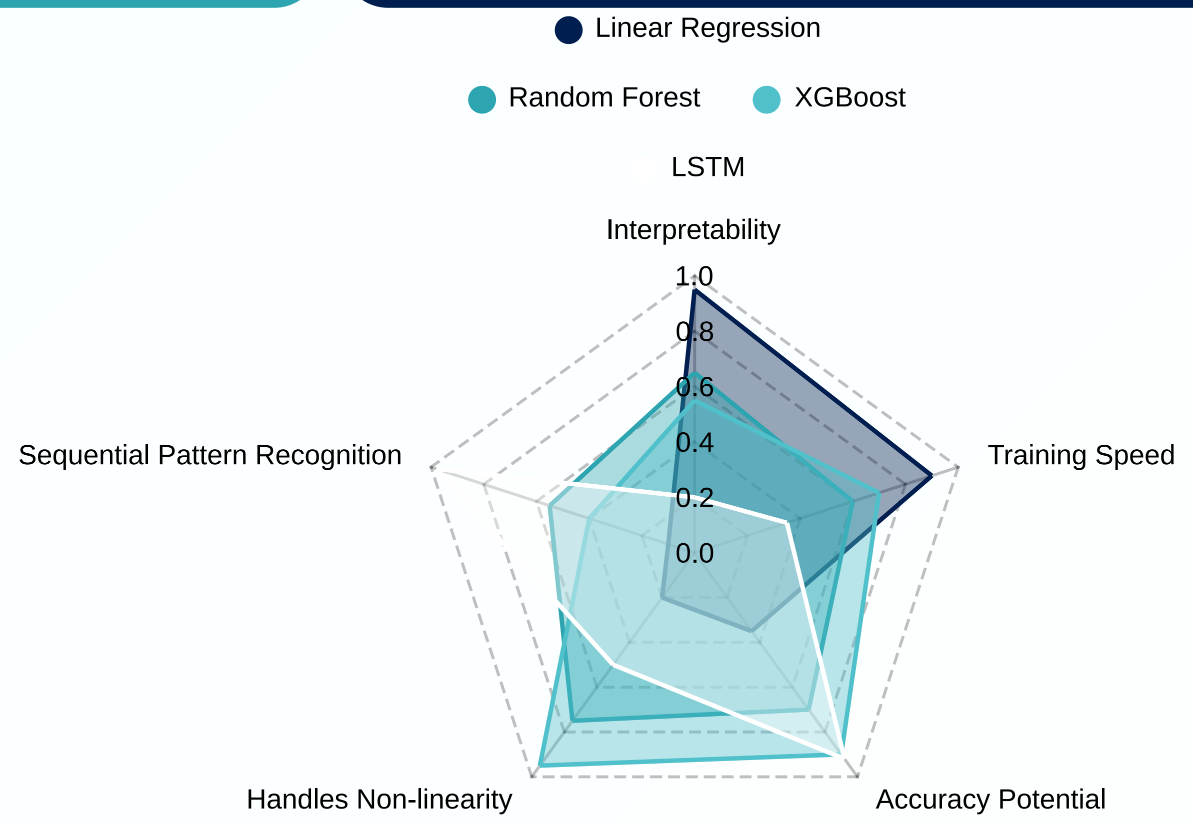
Ensemble method using multiple decision trees to capture non-linear relationships in stock data. Handles feature interactions effectively and provides feature importance rankings. Robust to outliers and reduces overfitting through averaging predictions across trees.

1. **XGBoost & LSTM**

XGBoost: Gradient boosting algorithm offering superior performance through sequential error correction. Highly efficient with built-in regularization. LSTM/GRU: Deep learning models designed for sequential time series data, capturing long-term dependencies in price movements.

Models Used

Four supervised machine learning algorithms selected for their complementary strengths in capturing linear relationships, complex patterns, and sequential dependencies in stock price data.



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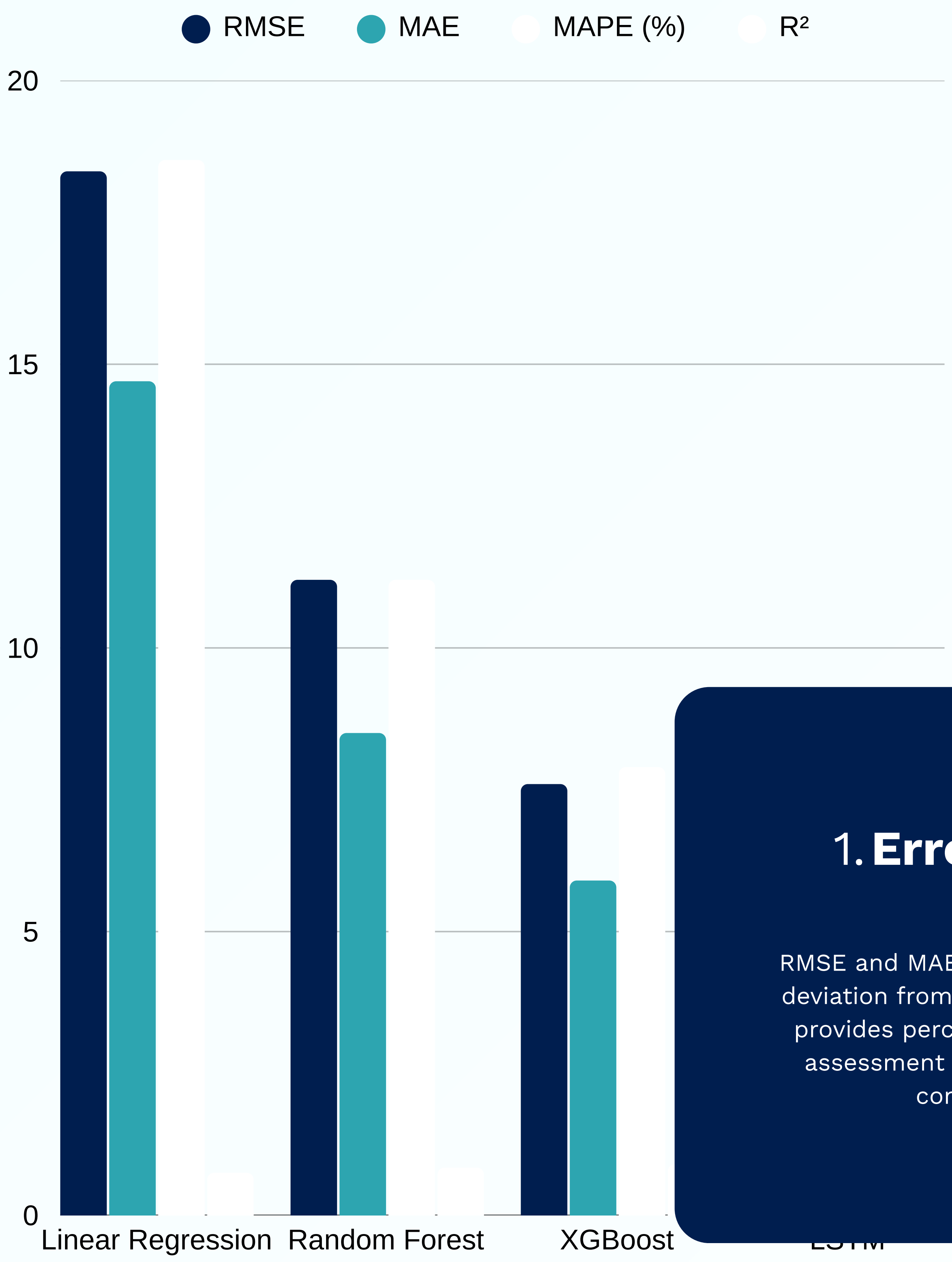

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Model Evaluation

Performance Metrics

Comprehensive evaluation using industry-standard metrics to assess prediction accuracy and model reliability. Backtesting performed with historical data splits to validate model performance across different market conditions.

1. Error Metrics

RMSE and MAE measure prediction deviation from actual prices. MAPE provides percentage-based error assessment for intuitive model comparison.

1. R^2 Score

Coefficient of Determination indicates variance explained by the model. Higher R^2 values demonstrate stronger predictive capability.

1. Directional

Directional accuracy measures correct prediction of price movement direction—crucial for trading decisions and investment timing.

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Deployment & Visualization

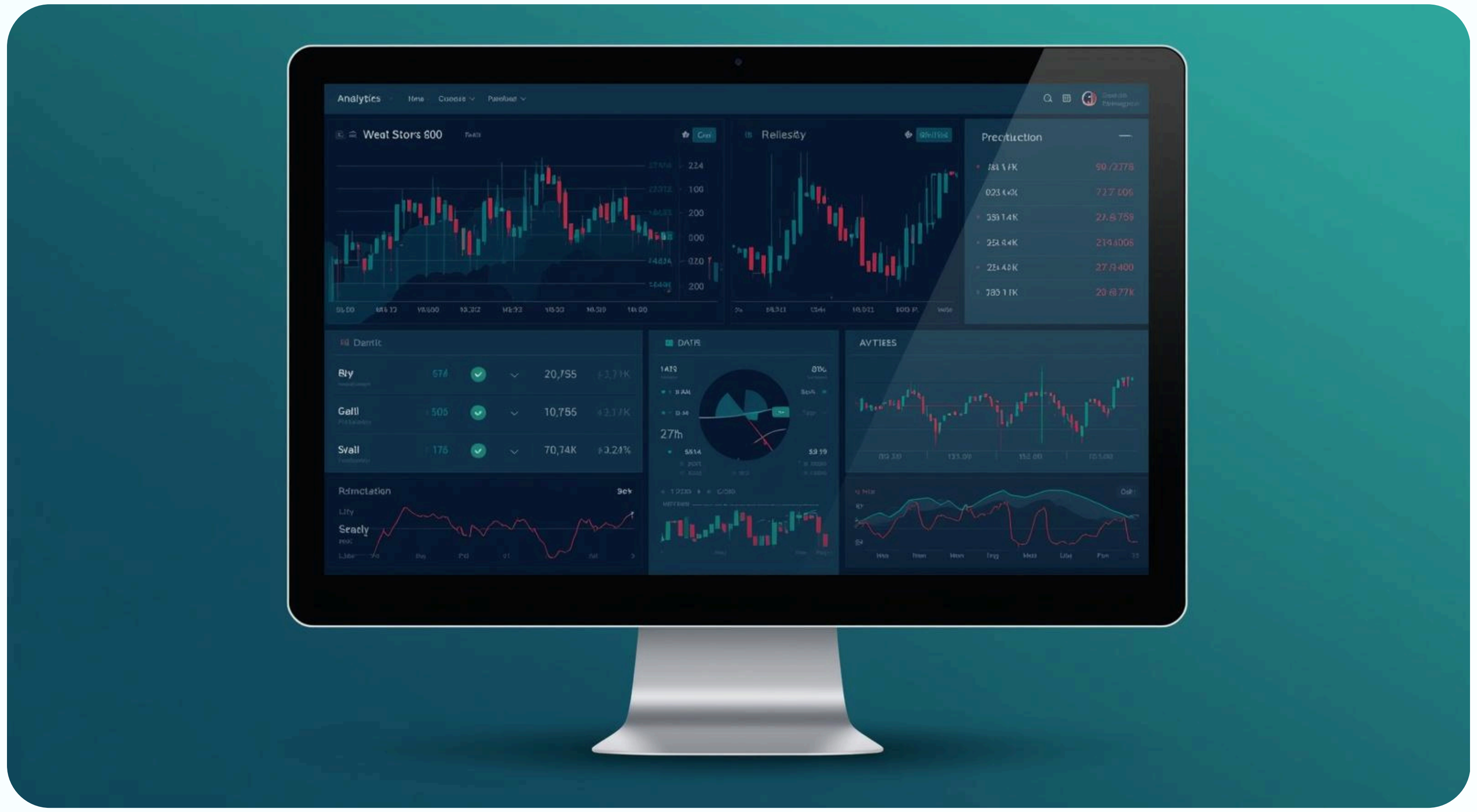
Interactive dashboard prototype built with Streamlit or Dash framework, enabling real-time stock price predictions and intuitive decision support for investors.

Dashboard Interface

User-friendly interface featuring interactive charts, prediction confidence intervals, and historical performance metrics. Designed for both technical analysts and everyday investors seeking data-driven insights.

Decision Support

Comprehensive toolkit combining price forecasts, trend analysis, and risk indicators. Empowers users to make informed investment decisions with clear visualizations and actionable recommendations.



Real-Time Updates

Automatic prediction refresh as new market data flows in. The system continuously processes latest Safaricom stock information to deliver up-to-date forecasts without manual intervention.

Buy/Sell Signals

Visual indicators clearly display recommended trading actions based on model predictions. Color-coded signals help investors quickly identify optimal entry and exit points for positions.

1. **Investment Decisions**

Enables improved, data-driven investment decisions by providing accurate stock price forecasts. Investors can leverage predictive insights to time their market entries and exits more effectively, reducing guesswork and emotional trading while maximizing potential returns on Safaricom equity investments.

1. **Risk Management**

Enhances risk management through predictive insights that identify potential market downturns before they occur. Our model's volatility indicators and trend analysis help investors set appropriate stop-loss levels, hedge positions strategically, and protect their portfolios from unexpected market movements.

1. **Portfolio Optimization**

Supports portfolio optimization with timely forecasts that inform asset allocation strategies. By providing localized market insights specific to Kenyan equities, our solution helps investors balance risk and reward, diversify holdings intelligently, and achieve better long-term financial outcomes.

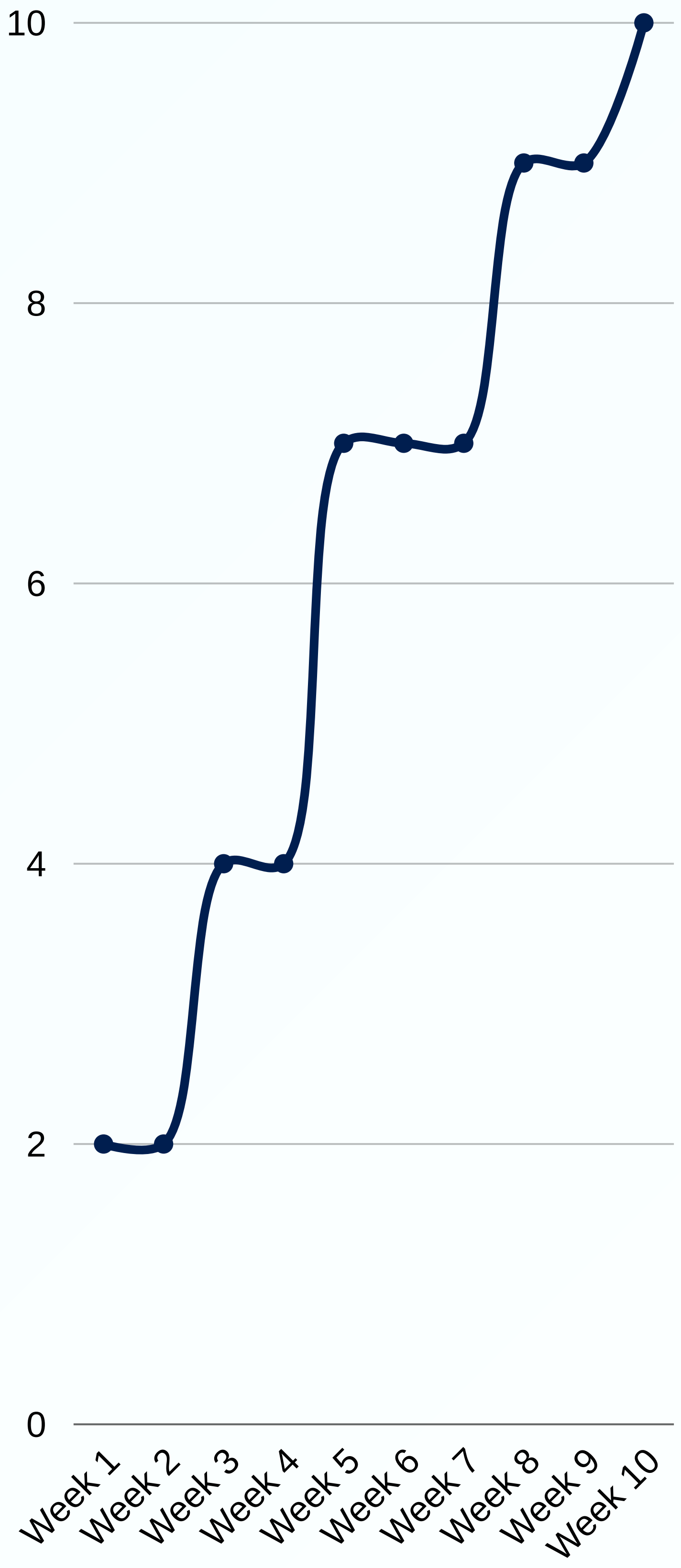
Business Impact

Our ML-powered stock prediction model delivers tangible value for investors and financial stakeholders in the Kenyan market, transforming raw data into actionable intelligence for smarter financial decisions.

Project
Timeline

Weeks 1–2: Data acquisition and preprocessing
Weeks 3–4: Feature engineering and EDA
Weeks 5–7: Model training and evaluation

Weeks 8–9: Dashboard development and deployment
Week 10: Final testing and report preparation



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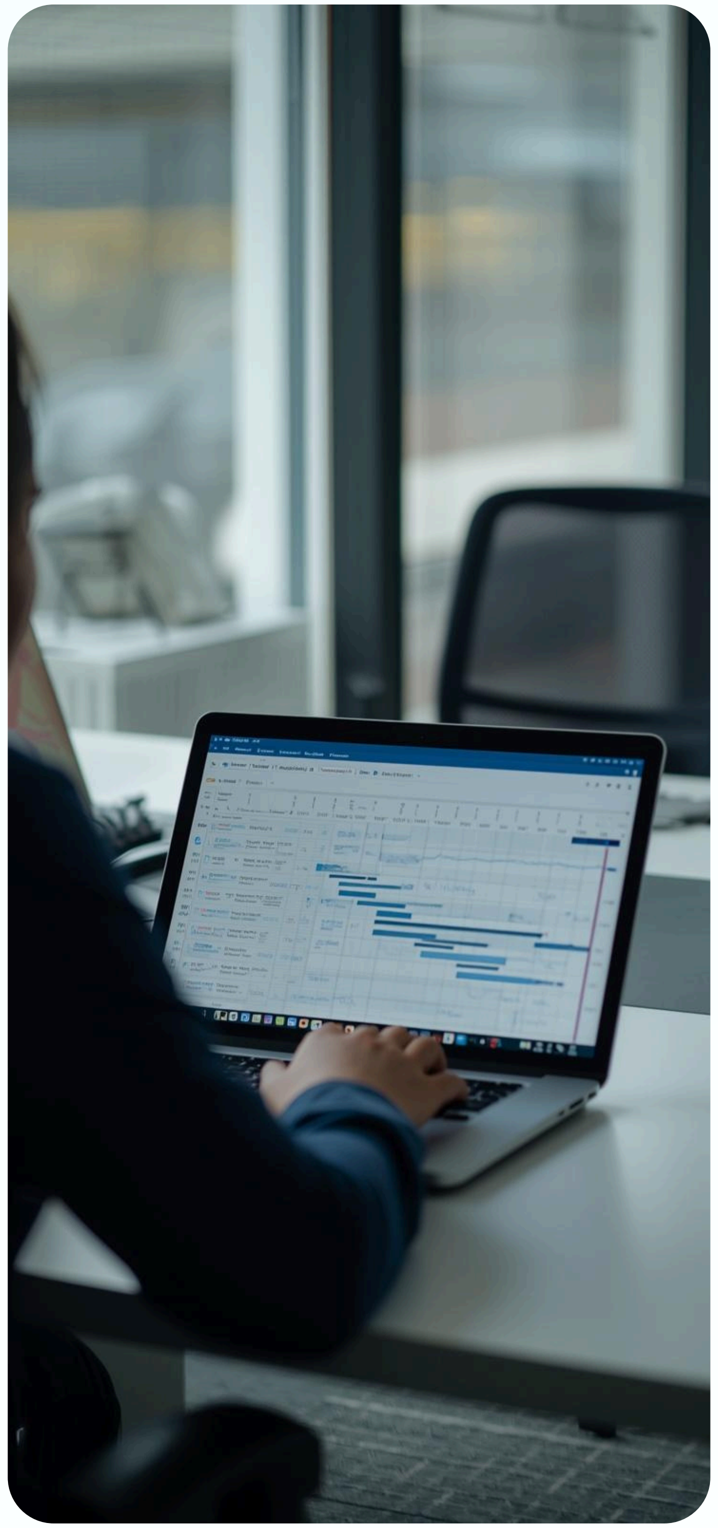

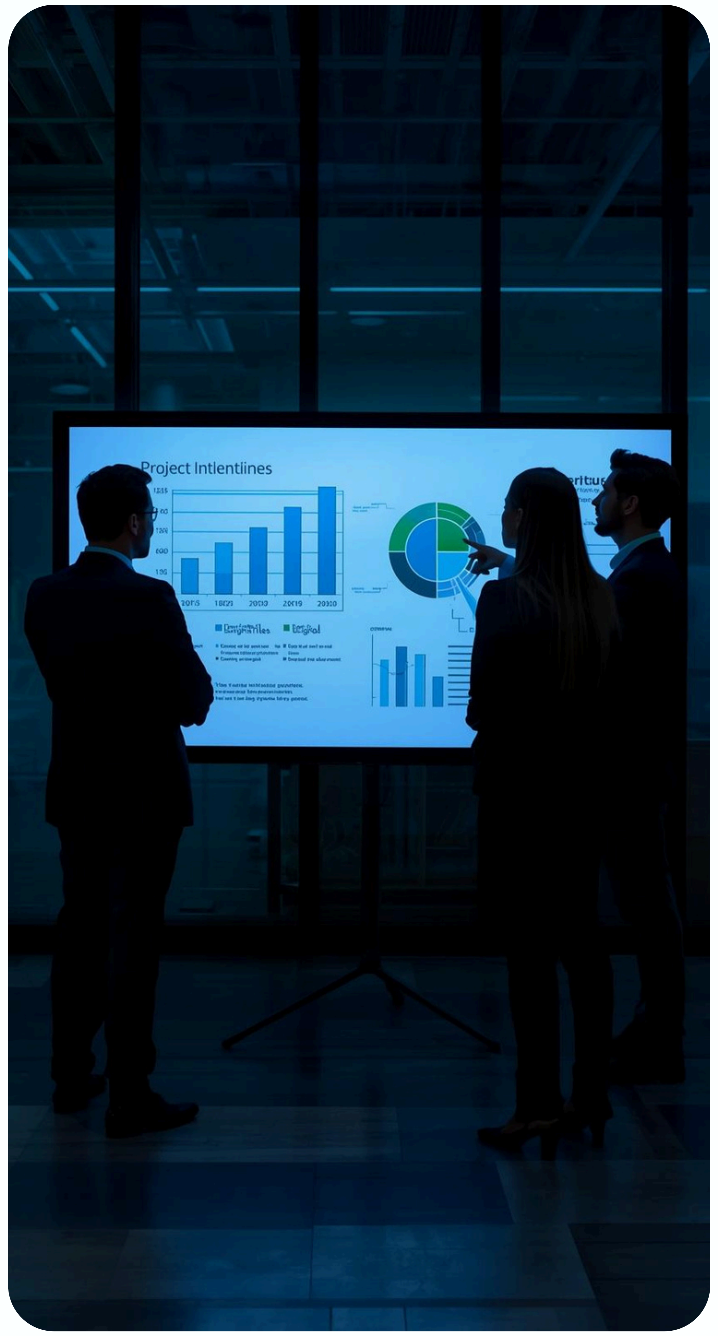

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Questions Welcome

Thank You

Questions are welcome. For further collaboration or inquiries about this stock price prediction project, please feel free to reach out.